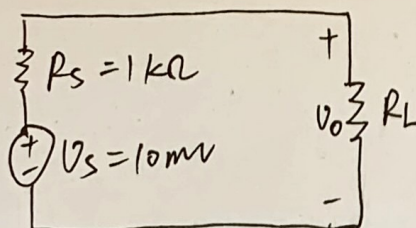




1.3

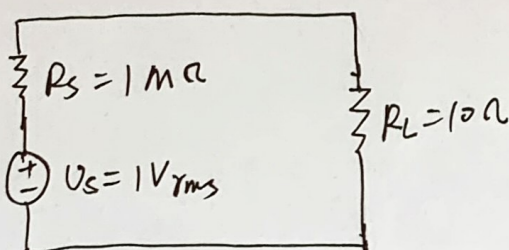


$$V_o = V_s \cdot \frac{R_L}{R_s + R_L}$$

① 当 $R_L = 100k\Omega, 10k\Omega, 1k\Omega, 100\Omega$ 时, 对应的 V_o 为
9.9mV, 9.1mV, 5mV, 0.9mV

② $\frac{R_L}{R_s + R_L} > 80\% \Rightarrow R_L > 4k\Omega$

1.12

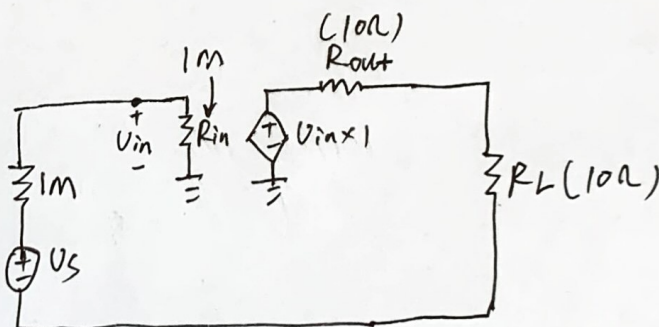
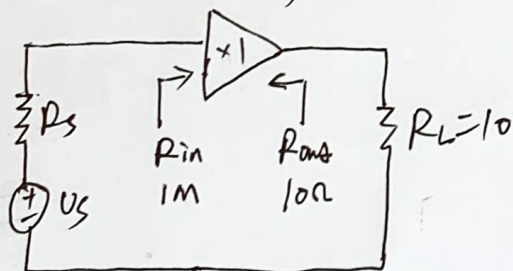


① connect directly:

$$V_L = 1 \cdot \frac{10}{1m + 10} = 10 \mu V_{rms}$$

$$P_L = \frac{V_L^2}{R_L} = 10^{-11} W$$

② insert a unity-gain buffer:



$$V_L = V_s \cdot \frac{1}{2} \cdot \frac{1}{2} = 0.25 V_{rms}$$

$$P_L = \frac{V_L^2}{R_L} = \frac{(0.25)^2}{10} = 6.25 mW$$

$$\frac{V_L}{V_s} = \frac{1}{4} = -12.04 dB$$

$$P_{in} = \frac{V_{in}^2}{R_{in}} = \frac{(0.5)^2}{1m} = 0.25 \mu W$$

$$\frac{P_L}{P_{in}} = \frac{6.25 mW}{0.25 \mu W} = 43.98 dB$$



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$$f = 10 \text{ Hz} : |H(j\omega)| = 60 \text{ dB}$$

$$f = 10 \text{ kHz} : 60 - 20 = 40 \text{ dB}$$

$$f = 100 \text{ kHz} : 20 \text{ dB}$$

$$f = 1 \text{ MHz} : 0 \text{ dB}$$

D.124

$$\frac{U_2}{U_S} = \frac{9 \text{ k}}{10 \text{ k} + \frac{1}{j\omega C}} = \frac{\frac{9 \text{ k}}{10 \text{ k}}}{1 + \frac{1}{j\omega C \cdot 10 \text{ k}}} \Rightarrow \text{高通}$$

$$\frac{1}{C \cdot 10 \text{ k}} \leq 2\pi \cdot 100 \text{ Hz} \Rightarrow C \geq 0.159 \mu\text{F}$$